

HEDGE: Enhancing Adversarial Resilience in Heterogeneous Intrusion Detection Ensembles via Directional Transfer-Aware Training

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Heterogeneous intrusion-detection ensembles combine neural and tree-based learners to improve predictive performance, yet architectural diversity does not guarantee adversarial independence. Adversarial examples crafted against one component may transfer to others and induce correlated failures, undermining the security benefits of ensemble learning. Existing defenses typically harden components in isolation or aggregate adversarial examples without explicitly modeling transfer direction. We present HEDGE, a defense framework for heterogeneous intrusion-detection ensembles based on Directional Transfer-Aware Training (DTAT). DTAT models adversarial exposure as a source-attack-target relationship and routes both direct and transferred adversarial examples into target-specific training curricula. This design explicitly incorporates cross-model transfer information during adversarial training rather than treating adversarial examples as an undifferentiated pool. We evaluate HEDGE using one deep neural network and four tree-based learners on the NSL-KDD and CSE-CIC-IDS2018 datasets under seven gradient- and query-based attacks. Common-input evaluation reveals strongly attack-dependent directional transfer, with DNN-originated DeepFool examples transferring substantially to multiple tree-based targets. Under the component-conditioned stress diagnostic, DTAT reduces DeepFool attack success rates relative to isolated adversarial training from 17.52% to 1.43% on NSL-KDD and from 6.43% to 2.03% on CSE-CIC-IDS2018 while preserving clean accuracies of 99.46% and 95.95%, respectively. These findings suggest that explicitly modeling directional adversarial exposure can improve empirical robustness in heterogeneous intrusion-detection ensembles under the evaluated threat model.

CCS Concepts: • **Security and privacy** → **Intrusion/anomaly detection and malware mitigation**; *Network security*; • **Computing methodologies** → *Machine learning*.

Additional Key Words and Phrases: adversarial machine learning, network intrusion detection, adversarial transferability, heterogeneous ensembles, adversarial training, empirical robustness

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1 Introduction

Machine-learning-based intrusion detection systems (IDSs) increasingly combine neural networks with tree-based learners to exploit complementary inductive biases on structured network-flow data [4, 12, 38, 39, 42]. Neural models provide flexible nonlinear representations, whereas boosting- and bagging-based tree ensembles remain effective on tabular features. Although this heterogeneity can improve clean-data performance and reduce dependence on an individual learner, it does not guarantee robustness when the shared input representation is deliberately manipulated. Adversarial perturbations can alter an initially correct security decision, either concealing malicious activity or assigning an intrusion to an incorrect response category [11, 14, 32, 45].

A central concern is *adversarial transferability*: an example generated against one model may remain effective against another without target-side re-optimization [20, 46, 47]. Transferability weakens the security rationale commonly attributed to model diversity. Components that implement different decision functions still process the same flow representation and may depend on overlapping or correlated features; a perturbation that crosses several component boundaries can therefore induce correlated errors that weighted voting cannot correct. The relevant unit of analysis is consequently not only the robustness of each component, but also the directed relation between the model used to generate an adversarial example, the attack procedure, and the model subsequently exposed to it.

This relation is especially consequential in neural–tree ensembles because the component attack surfaces differ. A differentiable DNN permits efficient gradient-based optimization, whereas split-based tree learners are commonly attacked through score or decision queries, structural search, or transferred examples. The two model families nevertheless share the same post-featurization input. A DNN-originated perturbation may therefore cross thresholds used by several tree components, while a perturbation optimized near a tree boundary may interact differently with the DNN loss surface. Transfer should thus be studied as an ordered source–attack–target relation rather than summarized as a symmetric ensemble property [1, 20].

Existing defenses address only parts of this problem. Conventional adversarial training primarily exposes a learner to examples generated against that learner [13, 23]; ensemble adversarial training and later IDS defenses show that externally generated or multi-source examples can improve empirical robustness [15, 21, 31, 37]. Other approaches exploit ensemble diversity, dynamic selection, attack detection, or input transformations [16, 18, 19, 22, 41, 43]. These mechanisms are valuable, but they do not by themselves define which source-generated perturbations should harden which deployed component. Moreover, adversarial IDS evaluations often blur direct attacks, transfer of an unchanged input, and diagnostics in which different ensemble branches receive separately optimized examples. Such conflation obstructs both causal attribution and operational interpretation [1, 2, 9].

The research problem addressed here is therefore narrower than generic adversarial training: *how should heterogeneous ensemble components be hardened against directed incoming adversarial exposure?* A defensible solution must preserve the identities of the generation source, attack mechanism, and protected target; construct target-specific curricula rather than an undifferentiated adversarial pool; and evaluate the resulting defense under a contract that separates direct attacks, unchanged common-input transfer, and stronger diagnostic stress conditions.

Accordingly, we investigate three complementary questions in a heterogeneous neural–tree IDS operating on post-featurization network-flow representations: how direct vulnerability and unchanged transfer vary across source–attack–target pathways; whether incorporating selected incoming DNN-to-tree exposure improves robustness beyond otherwise matched component-isolated adversarial training; and whether the resulting ensemble preserves clean predictive utility while maintaining the supporting role of inference-time hardening mechanisms.

We propose HEDGE, a heterogeneous ensemble defense whose core mechanism is Directional Transfer-Aware Training (DTAT). DTAT indexes each adversarial pool by its generation source and selectively assigns that pool to a protected target. In the evaluated neural-tree instantiation, a tree component is trained on query-based examples generated directly against it and on gradient-derived examples generated against the DNN source and supplied unchanged to the tree target. The DNN is trained on gradient- and query-based examples generated directly against itself. This policy prioritizes the incoming DNN-to-tree pathway specified by the mixed-access threat model instead of imposing symmetric all-to-all pooling. HEDGE retains component-specific feature quantization and fixed weighted voting as supporting mechanisms; the methodological contribution lies in the threat-to-curriculum mapping implemented by DTAT.

The contributions are threefold:

- **Directed evaluation model.** We formulate heterogeneous adversarial exposure as a source-attack-target relation and distinguish direct attacks, unchanged common-input transfer, and a component-conditioned ensemble stress diagnostic. We also separate multiclass decision disruption from strict malicious-to-benign bypass, preventing a generic classification error from being overinterpreted as successful detection evasion.
- **Directional Transfer-Aware Training.** We introduce DTAT, which maps direct and cross-family adversarial pools to target-specific curricula. The construction enables non-differentiable tree learners to learn from gradient-derived DNN examples while retaining direct query-based coverage, and it exposes a controlled difference from isolated adversarial training.
- **Controlled empirical evidence.** Across two IDS datasets, five heterogeneous learners, and seven attacks, we characterize direct vulnerability and common-input transfer, then isolate the contribution of DTAT through an IAT-DTAT ablation. The results show that the additional benefit is concentrated on pathways represented in the transferred curriculum, whereas feature squeezing provides a smaller and non-uniform increment.

The remainder of this article is organized as follows. Section 2 reviews adversarial evasion, transfer, and defenses for heterogeneous IDSs. Section 3 defines the ensemble, security objective, directed transfer relation, evaluation contract, and metrics. Section 4 presents HEDGE and DTAT. Section 5 reports the experimental evaluation and discussion. Section 6 concludes the article.

2 Related Work

2.1 Adversarial Evasion and Transfer in IDSs

Adversarial evasion against an IDS is governed jointly by the classifier, attacker knowledge, mutable traffic attributes, and the mapping between network behavior and extracted features. Gradient-based attacks such as FGSM, PGD, JSMA, DeepFool, and C&W expose local or optimization-based vulnerabilities of differentiable models [5, 13, 23, 24, 28]; ZOO and HSJA relax analytical-gradient access through score or decision queries [6, 7]. IDS-specific studies show that both classes of attacks can substantially degrade learned detectors, but effectiveness varies with the data representation, attack budget, mutable-feature policy, and evaluation denominator [8, 14, 40, 45].

Network-flow features are not freely interchangeable with image pixels. They include immutable identifiers, categorical fields, counters, timing variables, and derived quantities linked by semantic or causal constraints. Range clipping and an L_p bound establish only limited feature-domain validity; they do not prove that a vector can be reconstructed as protocol-valid traffic while preserving malicious functionality [1, 9, 14]. Accordingly, a feature-space experiment can diagnose vulnerabilities at the classifier boundary, but its conclusions must not be presented as packet-level realizability.

Transferability adds a second dimension. A source-generated example can affect a different target because the models share predictive structure, even when their architectures differ [20]. In IDSs, transferable attacks have been demonstrated across classifiers, autoencoder detectors, and flow representations [46, 47]. Recent attacks also constrain perturbations through feature importance or domain-specific projections, underscoring that transfer depends on the source, attack construction, target, and feasible feature set rather than on architecture alone [8, 40]. These observations motivate a directed source–attack–target treatment of transfer.

2.2 Defenses for Heterogeneous IDSs

Adversarial training remains the most established empirical defense. Standard formulations optimize a model over adversarial neighborhoods generated from its own loss or predictions [13, 23]. Ensemble adversarial training showed that examples generated by other models can reduce black-box vulnerability [37]; subsequent IDS systems combine adversarial retraining with ensemble learning, multiple attack sources, or model selection [3, 15, 21, 31]. These approaches establish that transferred or heterogeneous adversarial data can be useful, but most either train a single target on a pooled set or treat the ensemble as the unit of aggregation rather than explicitly routing source-indexed examples to distinct protected targets.

Heterogeneous ensemble defenses pursue complementary strategies. ELAT combines adversarial training and classifiers drawn from several model families [21]. ProvADShield measures mutual transferability to select provenance-based detector pairs with lower shared exposure [22]. More recent systems use dynamic model selection or multi-model voting to adapt the active detector set under attack [16, 19]. These works use heterogeneity to diversify or select decisions; HEDGE instead uses a directed transfer hypothesis to construct the training curriculum of each deployed component.

Input transformations and attack detectors provide additional layers. Feature squeezing maps nearby numerical inputs to a coarser representation [43]; autoencoder- and uncertainty-based detectors seek to reject adversarial inputs before or alongside classification [18, 30, 41]. Such mechanisms can complement robust training, but non-smooth preprocessing can also obscure gradients and must be evaluated against attacks adapted to the complete pipeline [2]. For this reason, HEDGE treats quantization as a secondary inference mechanism rather than evidence of robustness by itself.

2.3 Research Gap and Positioning of HEDGE

HEDGE builds on, but is not synonymous with, prior ensemble adversarial training. Tramèr et al. use adversarial examples from external models to harden a target classifier [37]; ELAT combines adversarial training and heterogeneous IDS ensemble selection [21]; Al Roken et al. aggregate white- and black-box examples for hybrid retraining of a target NIDS [31]; and ProvADShield uses directed mutual transferability to choose ensemble members [22]. The distinction introduced here is a *target-specific routing policy*: each generated pool retains its source and attack identity and is assigned only to the target curriculum associated with the prioritized incoming pathway.

The evaluated architecture also descends from APELID, which introduced training-only class balancing and parallel neural–tree inference [39]. APELID+ and APELID-ATH subsequently combined adversarial training, feature squeezing, and the same general ensemble lineage [10, 26]. HEDGE does not claim these elements as new. Relative to that preliminary work, the present study contributes the explicit source–attack–target formulation, the distinction between common-input transfer and component-conditioned stress, target-specific directional curricula, and a controlled isolated-versus-directional training ablation across two datasets.

Table 1. Positioning of HEDGE relative to representative defenses. A check mark denotes an explicit property reported by the cited method.

Approach	Heterogeneous deployed en- semble	Directed trans- fer analysis	Source-indexed target-specific training	Transfer- informed infer- ence/selection
Ensemble adversarial training [37]	–	Partial	–	–
ELAT [21]	✓	✓	–	✓
Hybrid adversarial retraining [31]	–	Partial	–	–
ProvADShield [22]	✓	✓	–	✓
APELID+/APELID-ATH [10, 26]	✓	Partial	–	–
HEDGE	✓	✓	✓	–

Table 1 summarizes this positioning. The novelty claim is intentionally narrow: HEDGE does not introduce transferred adversarial training in general; it operationalizes directed cross-family exposure as a component-specific curriculum policy for a deployed heterogeneous IDS ensemble.

3 Threat Model and Security Metrics

3.1 Heterogeneous Ensemble

Let $\mathcal{D} = \{(x_j, y_j)\}_{j=1}^N$ be a network-flow dataset, where $x_j \in \mathcal{X} \subseteq \mathbb{R}^d$ is the normalized feature vector produced by a fixed flow-featurization pipeline and $y_j \in \mathcal{Y} = \{0, \dots, m-1\}$ is its ground-truth class. Class 0 denotes benign traffic and the remaining classes denote intrusion categories.

The IDS is a heterogeneous ensemble $\mathcal{E} = \mathcal{T} \cup \mathcal{N}$ comprising tree-based learners $\mathcal{T} = \{T_1, \dots, T_k\}$ and neural learners $\mathcal{N} = \{N_{k+1}, \dots, N_n\}$. The evaluated instantiation contains XGBoost (XGB), Gradient Boosting Machine (GBM), CatBoost (CBT), a bagging meta-estimator (BME), and one DNN. We use *tree-based* rather than GBT when referring to the complete family because BME is bagging-based.

For an input x , component $M_i \in \mathcal{E}$ produces a normalized class-score vector $p_i(x)$. Fixed weighted soft voting gives

$$S_c(x) = \sum_{i=1}^n \omega_i p_{i,c}(x), \quad f_{\text{ens}}(x) = \arg \max_{c \in \mathcal{Y}} S_c(x), \quad (1)$$

where $\omega_i \geq 0$ and $\sum_i \omega_i = 1$.

3.2 Security Objective and Threat Model

Security semantics. For multiclass IDS output, an exact-label error and a detection bypass are not equivalent. Define the alarm mapping

$$b(c) = \begin{cases} 0, & c = 0, \\ 1, & c \neq 0. \end{cases} \quad (2)$$

For a malicious record $y \neq 0$, $f(x') \neq y$ is a multiclass integrity failure; strict bypass additionally requires $b(f(x')) = 0$. Unless stated otherwise, the reported attack-success rates measure conditional exact-label disruption. They do not treat every malicious-to-malicious transition as a successful avoidance of the alarm.

Adversarial position and objective. The adversary acts after flow featurization and before learned inference. Given clean input x , source model M_s , and attack a , the adversary produces $x'_{s,a} =$

$\Pi_{\mathcal{X}}(x + \delta_{s,a})$, where $\Pi_{\mathcal{X}}$ repairs the perturbed vector to the admissible feature domain. The primary objective is untargeted disruption of the source,

$$\delta_{s,a}^* \in \arg \max_{\delta \in \Delta_a(x)} \mathcal{L}(M_s(\Pi_{\mathcal{X}}(x + \delta)), y), \quad (3)$$

with an equivalent score- or decision-based objective when source gradients are unavailable.

Knowledge and access. For the DNN source, white-box access includes the architecture, parameters, preprocessing, and input gradients. Gradient-based attacks are therefore instantiated against the DNN. This is an experimental access choice, not a claim that structural white-box attacks on trees are impossible. Under black-box access, ZOO uses source class scores and HSJA uses source decisions; both can be instantiated against neural and tree-based sources [6, 7]. A transfer evaluation never uses target gradients or target-side re-optimization.

Feasible feature-space perturbations. Let $C \subseteq \{1, \dots, d\}$ denote mutable continuous coordinates and $\mathcal{I}$ their immutable, binary, categorical, or semantics-critical complement. For attack a ,

$$\begin{aligned} \Delta_a(x) &= \{\delta \in \mathbb{R}^d : \delta_r = 0 \ \forall r \in \mathcal{I}; \\ &\quad \|\delta_C\|_{p_a} \leq \epsilon_a \text{ when specified}; \\ &\quad \Pi_{\mathcal{X}}(x + \delta) \in \mathcal{X}\}. \end{aligned} \quad (4)$$

The repair operator performs inverse normalization, range clipping, non-negativity enforcement, and renormalization where applicable. Query attacks are additionally bounded by their configured stopping and iteration rules. These controls establish feature-domain validity, not packet-level realizability. As rigorously demonstrated by Cortellazzi et al. in their recent problem-space formalization [9] and Han et al. [14], mapping feature-space perturbations back to functional network traffic involves resolving complex inverse transformations, preserving protocol semantics, and maintaining the intrinsic malicious behavior. Addressing these strict problem-space constraints requires domain-specific generative frameworks that fall outside the scope of our current evaluation. Consequently, our threat model is deliberately bounded at the feature-extraction stage to isolate and analyze cross-model adversarial transferability, providing a worst-case feature-space diagnostic rather than a fully realizable packet-level attack [1].

Consequently, our threat model is deliberately bounded at the feature-extraction stage to isolate and analyze cross-model adversarial transferability, providing a controlled feature-space diagnostic rather than a fully realizable packet-level attack [1].

3.3 Directed Source–Attack–Target Transfer

Let s and t denote source and target models. For test item (x_j, y_j) , attack a is optimized against M_s to produce $x'_{j;s,a} = \Pi_{\mathcal{X}}(x_j + \delta_{j;s,a})$. The same repaired vector is supplied without modification to M_t . The evaluation is direct when $s = t$ and transferred when $s \neq t$. The source–attack–target triple (s, a, t) is retained in all labels and tables; a gradient-derived input evaluated on a tree is consequently reported as DNN-to-tree transfer, not as a direct white-box tree attack.

For attack a , the directed target-side transfer matrix is $\mathbf{T}_{s,t}^{(a)} = \text{ASR}_{s \rightarrow t}^{(a)}$. In general, $\mathbf{T}_{s,t}^{(a)} \neq \mathbf{T}_{t,s}^{(a)}$ because the generation procedure, source geometry, target vulnerability, and source–target boundary alignment are directional [20]. The present study evaluates operational non-reciprocity under a mixed-access attack suite; it does not infer a universal architecture-only ordering between neural and tree models.

3.4 Evaluation Contract and Security Metrics

The evaluation contract contains three distinct conditions.

- (1) **Direct component attack.** Attack a is generated against and returned to the same model, $s = t$.
- (2) **Common-input directed transfer.** Attack a is optimized only against source s and the unchanged $x'_{s,a}$ is evaluated on target $t \neq s$. This condition supports transferability claims.
- (3) **Component-conditioned ensemble stress diagnostic.** Tree and neural branches may receive separately generated adversarial representations of the same original record before score aggregation. This oracle-style condition diagnoses simultaneous family-specific failure, but it is not a realizable common-input attack because the branches can receive different vectors.

Let h denote a component or the ensemble and let $\mathcal{I}_h = \{j : h(x_j) = y_j\}$ be the initially correct indices. Clean accuracy and robust accuracy are

$$\text{Acc}_{\text{clean}}(h) = \frac{1}{N} \sum_{j=1}^N \mathbb{1}\{h(x_j) = y_j\}, \text{Acc}_{\text{rob}}^{(s,a)}(h) = \frac{1}{N} \sum_{j=1}^N \mathbb{1}\{h(x'_{j;s,a}) = y_j\}. \quad (5)$$

Attack success is conditioned on the evaluated target being initially correct:

$$\text{ASR}_{s \rightarrow h}^{(a)} = \frac{\sum_{j=1}^N \mathbb{1}\{h(x_j) = y_j \wedge h(x'_{j;s,a}) \neq y_j\}}{\sum_{j=1}^N \mathbb{1}\{h(x_j) = y_j\}}. \quad (6)$$

Equation (6) is not generally $1 - \text{Acc}_{\text{rob}}$, because robust accuracy includes examples that were already incorrect before perturbation. When $s \neq h$, we call it *directed target-side ASR*; it measures delivered target risk and does not require source-model failure. Precision, support-weighted F1, and recall are retained for comparability with IDS reporting. In a single-label multiclass setting, support-weighted recall equals accuracy and is not interpreted as an independent statistic.

4 HEDGE: Proposed Method for Heterogeneous Intrusion-Detection Ensembles

4.1 Framework Overview

The design objective of HEDGE is to harden heterogeneous ensemble components against adversarial exposure that originates from different model-attack pathways, while preserving the clean-data utility and deployment structure of the underlying IDS. Rather than treating adversarial examples as a single augmentation pool, HEDGE separates the generation source, attack mechanism, and protected target so that relevant incoming exposure can be incorporated into the curriculum of each component. This design makes directional transfer an explicit training consideration while retaining the existing heterogeneous ensemble as the deployment unit. [Figure 1](#) summarizes the resulting framework in three stages. The first stage establishes a controlled clean-data reference through training-only preprocessing and class balancing inherited from APELID [39]. The second stage implements the methodological core of HEDGE: DTAT generates source-indexed adversarial pools and selectively routes them to target-specific curricula according to the prescribed source-attack-target policy. The third stage performs hardened inference, where each component receives a deterministic, component-specific quantized view of the same deployment input and the resulting class scores are combined by fixed weighted voting. The first and third stages provide the data-preparation and inference context required for a complete IDS instantiation, whereas the defining methodological contribution lies in the directional assignment of adversarial exposure during training.

4.2 Design Motivation and Directional Transfer Mapping

The design of DTAT starts from the mixed-access threat model defined in [Section 3](#), rather than from a transfer ranking inferred from the test set. The evaluated ensemble combines a differentiable

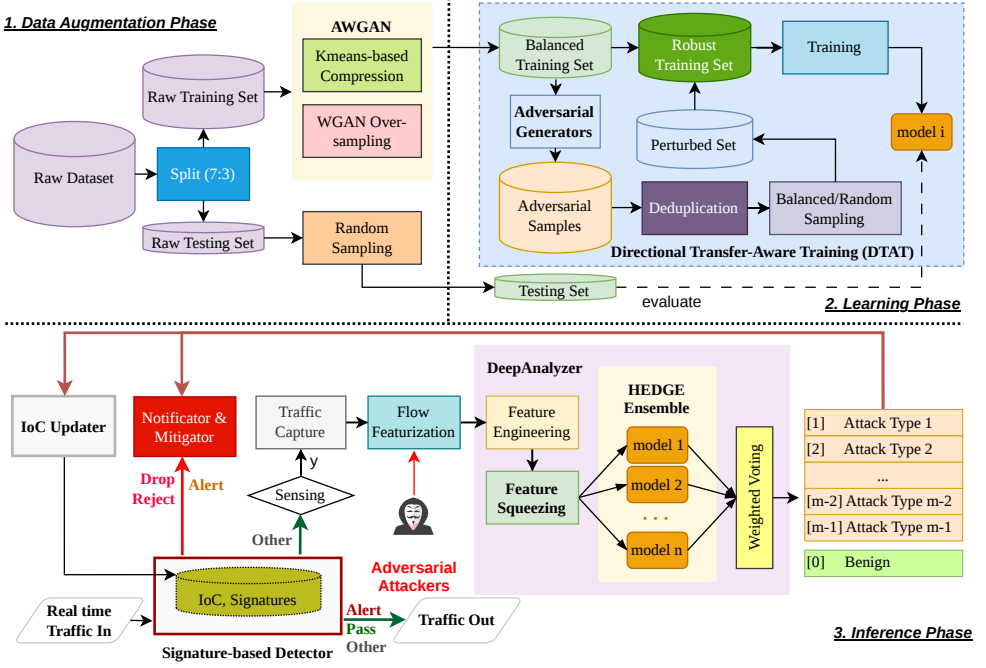


Fig. 1. The deployment context of the HEDGE framework. Phase 1 provides data augmentation. Phase 2 (the core contribution of this work) introduces Directional Transfer-Aware Training (DTAT), explicitly mapping source-indexed adversarial pools to target curricula. Phase 3 aggregates decisions via feature squeezing and weighted voting within the DeepAnalyzer engine. Components such as signature detectors and mitigators illustrate the broader operational pipeline but fall outside the scope of our adversarial transfer evaluation.

DNN with non-differentiable tree-based components operating on the same post-featurization representation. Under the considered access model, gradient-based attacks are generated against the DNN, whereas score- and decision-based attacks can be instantiated against both neural and tree sources. Because all components consume the same feature representation, adversarial examples generated against one component constitute a plausible incoming exposure for another component even when their attack-generation interfaces differ.

We represent such exposure as an ordered source–attack–target relation. Let $\Gamma \subseteq \mathcal{E} \times \mathcal{A} \times \mathcal{E}$ denote the pathways included in the training policy, where $(s, a, t) \in \Gamma$ indicates that examples generated by attack a against source s are eligible for the curriculum of protected target t . In the evaluated configuration, Γ contains the direct tree pathways (T_i, a_{BB}, T_i) , the direct neural pathways (N, a, N) , and the cross-family pathways (N, a_{WB}, T_i) . Tree-to-DNN transfer is evaluated experimentally but is not included as a default training pathway.

Figure 2 summarizes the corresponding attack-generation and cross-model exposure structure. Gradient-based attacks (FGSM, JSMA, PGD, DeepFool, and C&W) originate from the differentiable DNN source, while the query-based attacks ZOO and HSJA can originate from either the DNN or a tree-based component. The generated examples are then evaluated either on their source model or, without target-side re-optimization, on another component to characterize directed transfer. The figure therefore distinguishes the broader set of pathways considered in the evaluation from the

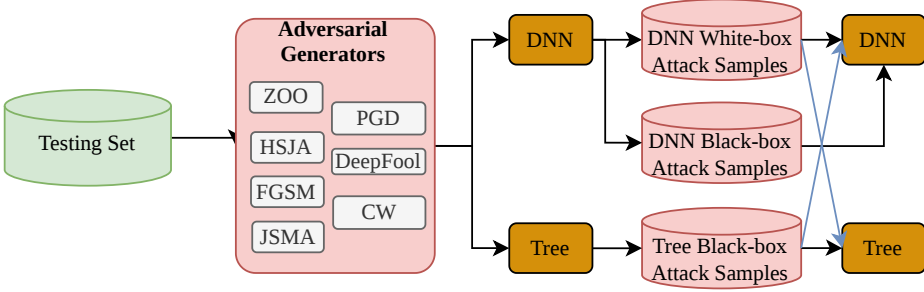


Fig. 2. Directional adversarial pathways evaluated in HEDGE. Gradient-based attacks originate from the DNN, whereas ZOO and HSJA use both neural and tree sources. Generated examples are evaluated directly or transferred unchanged across components; DTAT selects a threat-model-relevant subset of these pathways for target-specific training.

subset Γ selected for DTAT training; in particular, tree-to-DNN query-based transfer is measured for analysis, whereas the additional cross-family training pathway is DNN-to-tree.

This mapping is a prespecified exposure policy rather than a claim that DNN-to-tree transfer is universally stronger than the reverse direction. DTAT does not estimate a continuous transfer-risk score or optimize an allocation over all possible source–attack–target triples. Instead, it operationalizes a finite set of threat-model-relevant pathways by determining which source-indexed adversarial pools are assigned to each protected target. The value of the additional DNN-to-tree pathway is evaluated subsequently through common-input transfer analysis and the controlled IAT–DTAT comparison.

The distinction from indiscriminate pooling is material. A pooled curriculum discards the generation-source and protected-target identities of adversarial examples, making it difficult to attribute observed robustness to the amount of adversarial data, the attack mechanism, or the incoming exposure represented during training. By retaining pathway identity, DTAT enables target-specific curriculum construction and controlled pathway-level attribution.

4.3 Directional Transfer-Aware Training

4.3.1 Adversarial Training Pools. Let $\mathcal{A}_{WB}$ contain FGSM, JSMA, PGD, DeepFool, and C&W, and let $\mathcal{A}_{BB}$ contain ZOO and HSJA. Clean seed models $M_s^{(0)}$ are fitted on the balanced training set $\mathcal{D}_{bal}$. For a source $M_s^{(0)}$, attack a , and training example (x, y) , generation produces $\tilde{x}_{s,a} = \Pi_X(x + \delta_{s,a})$, while retaining ground-truth label y .

For the evaluated architecture, the source-indexed pools are

$$A_N^{WB} = \{(\tilde{x}_{N,a}, y) : a \in \mathcal{A}_{WB}, (x, y) \in \mathcal{D}_{bal}\}, \quad (7)$$

$$A_{T_i}^{BB} = \{(\tilde{x}_{T_i,a}, y) : a \in \mathcal{A}_{BB}, (x, y) \in \mathcal{D}_{bal}\}, \quad (8)$$

$$A_N^{BB} = \{(\tilde{x}_{N,a}, y) : a \in \mathcal{A}_{BB}, (x, y) \in \mathcal{D}_{bal}\}. \quad (9)$$

Generated examples are repaired, deduplicated, and sampled to preserve attack and class coverage. Let $\text{Curate}_\rho(A; \mathcal{D}_{bal})$ denote this operation, with at most $\rho|\mathcal{D}_{bal}|$ adversarial examples. The experiments use $\rho \approx 0.5$, preventing the adversarial portion from overwhelming the clean reference distribution.

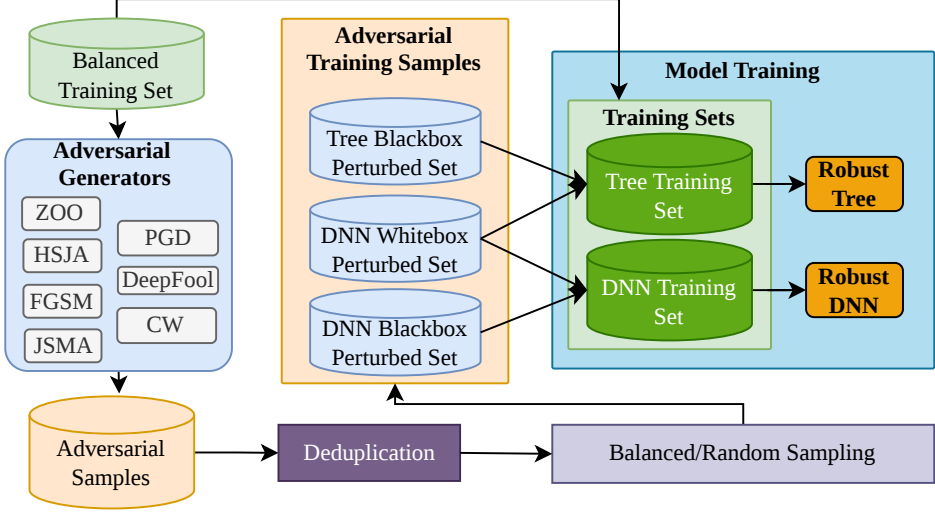


Fig. 3. Target-specific curriculum construction under DTAT. Source-indexed adversarial pools are selectively routed to protected targets: tree curricula combine direct query-based and DNN-originated gradient-based examples, whereas the DNN curriculum contains adversarial examples generated directly against the DNN.

4.3.2 *Target-Specific Curriculum Construction.* For a tree target T_i , DTAT constructs

$$\mathcal{D}_{T_i}^{\text{DTAT}} = \mathcal{D}_{\text{bal}} \cup \text{Curate}_\rho \left(A_{T_i}^{\text{BB}} \cup A_N^{\text{WB}}; \mathcal{D}_{\text{bal}} \right). \quad (10)$$

The first pool provides direct query-based coverage through adversarial examples generated against T_i itself. The second pool represents the selected incoming cross-family pathway: gradient-based adversarial examples generated against the DNN source are supplied unchanged to the tree target during retraining. Hence, the tree curriculum combines direct target-specific exposure with source-indexed transferred exposure while preserving the origin of each adversarial pool.

For the neural target, DTAT constructs

$$\mathcal{D}_N^{\text{DTAT}} = \mathcal{D}_{\text{bal}} \cup \text{Curate}_\rho \left(A_N^{\text{WB}} \cup A_N^{\text{BB}}; \mathcal{D}_{\text{bal}} \right). \quad (11)$$

This curriculum covers gradient- and query-based adversarial examples generated directly against the DNN. Tree-originated examples are not included in the neural curriculum under the evaluated policy. This omission does not imply that tree-to-DNN transfer is negligible; rather, the present design isolates the additional DNN-to-tree pathway specified by Γ instead of implementing symmetric all-to-all adversarial augmentation.

Figure 3 illustrates the selective routing defined algebraically in Equations (10) and (11). Unlike an undifferentiated adversarial pool, the construction retains the generation source, attack mechanism, and protected target associated with each adversarial set. Consequently, tree targets are exposed during retraining to both their direct query-based adversarial examples and the selected DNN-originated gradient-based examples, while the neural target is trained on attacks generated directly against itself. The resulting curricula therefore implement the prespecified directional policy introduced in Section 4.2 without assuming that omitted transfer directions are universally weak or irrelevant.

Algorithm 1 Directional Transfer-Aware Training in HEDGE

Require: Raw data $\mathcal{D}_{\text{raw}}$; tree models $\mathcal{T}$; neural models $\mathcal{N}$; attacks $\mathcal{A}_{\text{WB}}, \mathcal{A}_{\text{BB}}$; budget ratio ρ

Ensure: Hardened models $\mathcal{T}^*, \mathcal{N}^*$

- 1: $\mathcal{D}_{\text{tr}}, \mathcal{D}_{\text{te}} \leftarrow \text{PreprocessSplit}(\mathcal{D}_{\text{raw}})$
- 2: $\mathcal{D}_{\text{bal}} \leftarrow \text{BalanceTrainingOnly}(\mathcal{D}_{\text{tr}})$
- 3: **for all** $M_i \in \mathcal{T} \cup \mathcal{N}$ **do**
- 4: $M_i^{(0)} \leftarrow \text{Train}(M_i, \mathcal{D}_{\text{bal}})$
- 5: **end for**
- 6: $A_N^{\text{WB}} \leftarrow \text{GeneratePool}(\mathcal{D}_{\text{bal}}, N^{(0)}, \mathcal{A}_{\text{WB}}, \Pi_X)$
- 7: $A_N^{\text{BB}} \leftarrow \text{GeneratePool}(\mathcal{D}_{\text{bal}}, N^{(0)}, \mathcal{A}_{\text{BB}}, \Pi_X)$
- 8: **for all** $T_i^{(0)} \in \mathcal{T}$ **do**
- 9: $A_{T_i}^{\text{BB}} \leftarrow \text{GeneratePool}(\mathcal{D}_{\text{bal}}, T_i^{(0)}, \mathcal{A}_{\text{BB}}, \Pi_X)$
- 10: $B_{T_i} \leftarrow \text{Curate}(A_{T_i}^{\text{BB}} \cup A_N^{\text{WB}}, \rho | \mathcal{D}_{\text{bal}}|)$
- 11: $T_i^* \leftarrow \text{Train}(T_i, \mathcal{D}_{\text{bal}} \cup B_{T_i})$
- 12: **end for**
- 13: $B_N \leftarrow \text{Curate}(A_N^{\text{WB}} \cup A_N^{\text{BB}}, \rho | \mathcal{D}_{\text{bal}}|)$
- 14: $N^* \leftarrow \text{Train}(N, \mathcal{D}_{\text{bal}} \cup B_N)$
- 15: **return** $\mathcal{T}^*, \mathcal{N}^*$

Following curriculum construction, each component is retrained by empirical risk minimization over its finite target-specific dataset:

$$\theta_i^* = \arg \min_{\theta_i} \frac{1}{|\mathcal{D}_i^{\text{DTAT}}|} \sum_{(z, y) \in \mathcal{D}_i^{\text{DTAT}}} \mathcal{L}(M_i(z; \theta_i), y). \quad (12)$$

Equation (12) is therefore an empirical-risk objective over a curated, attack-aware curriculum rather than an exact robust-optimization objective. In particular, DTAT does not solve an inner maximization over the complete feasible perturbation set, estimate robustness outside the represented attack pools, or provide a certified guarantee over the admissible feature domain.

4.3.3 Isolated vs. Directional Transfer-Aware Training. The isolated tree baseline excludes the incoming cross-family pool:

$$\mathcal{D}_{T_i}^{\text{IAT}} = \mathcal{D}_{\text{bal}} \cup \text{Curate}_{\rho}(A_{T_i}^{\text{BB}}; \mathcal{D}_{\text{bal}}). \quad (13)$$

The neural curriculum is unchanged because both IAT and DTAT train the DNN on its direct white- and black-box examples. Under the same clean data, models, attacks, ensemble weights, and adversarial budget, the inclusion of A_N^{WB} in Equation (10) is therefore the controlled methodological difference. The IAT-DTAT comparison tests whether DNN-originated exposure adds robustness beyond direct tree hardening.

4.3.4 Training Procedure. Algorithm 1 summarizes the complete training process. All adversarial training examples are generated from the training partition; the test partition is never used to construct curricula.

4.4 Inference Hardening

At inference, feature squeezing maps mutable normalized coordinates to a component-specific bit depth b_i :

$$[Q_{b_i}(x)]_r = \begin{cases} \lfloor \frac{x_r(2^{b_i} - 1)}{2^{b_i} - 1} \rfloor, & r \in C, \\ x_r, & r \in I. \end{cases} \quad (14)$$

Categorical, binary, and semantics-critical coordinates are left unchanged. Smaller b_i contracts more numerical variation but can remove predictive detail; the selected values are fixed before adversarial evaluation. Quantization is a deterministic transformation of the same incoming vector for every component, even when bit depths differ.

The final HEDGE decision is

$$f_{\text{HEDGE}}(x) = \arg \max_{c \in \mathcal{Y}} \sum_{i=1}^n \omega_i [M_i^*(Q_{b_i}(x))]_c. \quad (15)$$

The weights do not adapt to individual inputs. Feature squeezing and weighted voting are therefore treated as supporting inference mechanisms rather than as the primary source of adversarial robustness; their incremental effects are examined separately from DTAT in Section 5.

4.5 Methodological Analysis

Computational complexity. Let n be the number of balanced training records, d the feature dimension, Q_a the query or iteration cost of attack a , and n_{adv} the number of generated examples. Data balancing has the cost of the retained AWGAN and K-means pipeline; white-box generation is linear in $n_{\text{adv}}d$ for fixed optimization steps, while query-based generation scales as $O(Q_a n_{\text{adv}}d)$. Retraining scales with the curated augmented-set size, bounded by $(1 + \rho)n$ per target. Inference-time quantization costs $O(d)$ per component and is dominated by model evaluation. The principal cost is offline adversarial generation and retraining, not the directional assignment itself.

Rationale for directional neural-tree transfer. The method does not assume a universal geometric hierarchy. A plausible mechanism is that models sharing a flow representation exploit overlapping predictive dependencies, while their local boundaries respond differently to the same coordinated feature shift. DNN optimization may identify multivariate directions that also cross tree thresholds, whereas tree-local perturbations need not align with high-loss neural directions. Transfer literature supports source–target and attack dependence rather than architecture-only causality [20, 46]. HEDGE therefore treats the DNN-to-tree pathway as an empirically testable exposure hypothesis.

Adversarial coverage of DTAT. Compared with IAT, Equation (10) expands the finite tree training set from direct query examples to the union of direct and DNN-originated gradient examples. This increases coverage only for attacks and boundary regions represented by the generated pools. It does not imply robustness to unseen algorithms, budgets, or traffic-space transformations.

Scope of robustness claims. Low ASR under a fixed attack configuration can reflect finite attack budgets, optimization failure, preprocessing effects, or gradient obfuscation [2]. Feature squeezing is therefore treated only as a secondary inference mechanism. Because the present evaluation does not include an adaptive adversary that differentiates through the complete repair–quantization–ensemble pipeline using gradient-approximation techniques, we do not attribute the principal robustness claim to quantization or claim fully adaptive white-box security. The primary evidence for DTAT is instead the controlled IAT–DTAT comparison, which measures the observed effect of adding the selected DNN-originated adversarial pool while keeping the remaining experimental configuration matched. Accordingly, the results establish empirical robustness only under the

reported feature-space and attack-aware conditions. The component-conditioned protocol remains an upper-stress diagnostic rather than a realizable common-input deployment attack; no certified, packet-level, unseen-attack, or fully adaptive robustness guarantee is asserted.

5 Experiments and Evaluation

This section presents the experimental evaluation of HEDGE through three research questions that examine directional adversarial exposure, the contribution of transfer-aware training, and the behavior of the complete defended system:

- RQ1. Directional adversarial exposure.** How do direct attack vulnerability and unchanged cross-model transfer vary across source–attack–target pathways in a heterogeneous neural-tree intrusion-detection ensemble?
- RQ2. Contribution of directional transfer-aware training.** Does incorporating the selected incoming DNN-to-tree exposure through DTAT reduce observed adversarial vulnerability beyond otherwise matched component-isolated adversarial training?
- RQ3. System-level robustness and utility.** Does the complete HEDGE ensemble preserve clean predictive performance, and what additional system-level effects are observed when feature squeezing and fixed weighted voting are included under the evaluated threat model?

Together, the three research questions examine directional adversarial exposure (RQ1), the contribution of DTAT (RQ2), and the robustness–utility trade-off of the complete HEDGE system (RQ3). The reported results provide attack-aware empirical evidence under a fixed-split feature-space setting; they do not support claims of statistical generalization, robustness to unseen attack families, certified robustness, or fully adaptive end-to-end security.

5.1 Experimental Setup

Datasets and data preparation. We evaluate NSL-KDD [36] and CSE-CIC-IDS2018 [34]. NSL-KDD retains 41 features, comprising 34 numerical and seven categorical attributes. For CSE-CIC-IDS2018, timestamps, invalid counters, inconsistent window attributes, duplicates, and features that become constant after cleaning are removed, leaving 62 features. Each cleaned dataset is divided by a stratified 70:30 record-level split. Encoders, scaling parameters, and admissible feature ranges are fitted on the training partition and applied unchanged to the test partition.

Class balancing is applied only to training data. Following APELID [39], minority classes are augmented through a WGAN-based procedure and majority classes are represented through K-means-based selection, producing 14,000 training records per class. Test majority classes are capped at 6,000 records to control attack-generation cost, while minority classes remain unchanged. The resulting test frequencies are therefore an evaluation distribution rather than an estimate of operational prevalence. Table 2 reports the distributions.

Model configuration. The ensemble contains XGB, GBM, CBT, BME, and one DNN. The single DNN provides a controlled differentiable source, while four tree implementations expose variation within the non-differentiable family. Hyperparameters, fixed ensemble weights, and component-specific bit depths are reported in Table 3. All settings are held constant across corresponding undefended and defended configurations.

Adversarial attack configuration. The attack suite contains score- and decision-based ZOO and HSJA [6, 7], together with FGSM, JSMA, PGD, DeepFool, and C&W [5, 13, 23, 24, 28]. Gradient-based attacks are generated against the DNN; ZOO and HSJA are generated against both neural and tree sources. Table 4 records the explicitly configured parameters in the original experiment. A dash denotes a non-applicable field. The archived aggregate results do not retain total per-example query counts or every library default; this limitation is revisited in Section 5.5.

Table 2. Dataset distributions after cleaning, training-only balancing, and test-set capping.

Class	Cleaned	Balanced train	Test
<i>NSL-KDD</i>			
Benign	77,120	14,000	6,000
DoS	52,987	14,000	6,000
Probe	13,935	14,000	4,181
R2L	3,738	14,000	1,122
U2R	108	14,000	33
<i>CSE-CIC-IDS2018</i>			
Benign	8,037,443	14,000	6,000
DDoS-LOIC-HTTP	574,215	14,000	6,000
DDoS-HOIC	198,861	14,000	6,000
DoS-Hulk	145,199	14,000	6,000
Bot	144,277	14,000	6,000
Infiltration	94,859	14,000	6,000
SSH-Bruteforce	94,045	14,000	6,000
DoS-GoldenEye	41,392	14,000	6,000
DoS-Slowloris	9,712	14,000	2,914
Brute Force-Web	406	14,000	122
Brute Force-XSS	223	14,000	67
SQL Injection	84	14,000	26

Table 3. Principal model, feature-squeezing, and ensemble parameters.

Configuration	XGB	GBM	CBT	BME	DNN
Estimators	1200	–	–	30	–
Iterations/epochs	–	100	2000	–	40
Learning rate	0.08	0.10	0.08	–	0.001
Maximum depth	8	3	8	–	–
FS bits: NSL-KDD	6	8	8	8	2
FS bits: CSE-CIC-IDS2018	6	2	2	2	2
Ensemble weight	0.3	0.2	0.2	0.2	0.1

Adversarial training-set construction. Training and evaluation attacks use disjoint records. Curriculum examples are generated only from $\mathcal{D}_{\text{bal}}$, repaired, deduplicated, approximately balanced across labels and attacks, and capped at roughly 50% of the balanced clean set. The same seven attack families appear during training and evaluation. The reported robustness is therefore attack-aware; it is not zero-shot evidence for unseen attacks.

Evaluation metrics. We report clean and adversarial accuracy, support-weighted F1, precision, and the conditional ASR in Equation (6). For direct attacks, source and target coincide. For transfer, the table identifies the generation source and evaluated target. All ensemble attack rows use the component-conditioned stress diagnostic unless explicitly labeled common-input.

Table 4. Recorded attack parameters for NSL-KDD and CSE-CIC-IDS2018.

Dataset	Parameter	ZOO	HSJA	JSMA	FGSM	PGD	DeepFool	C&W
NSL-KDD	Confidence	0.5	–	–	–	–	–	0.0
NSL-KDD	ϵ	–	–	–	0.1	0.2	–	–
NSL-KDD	Learning rate	0.05	–	–	–	–	–	0.01
NSL-KDD	Step size	–	–	–	0.01	0.01	–	–
NSL-KDD	θ / γ	–	–	0.02 / 0.10	–	–	–	–
NSL-KDD	Maximum iterations	50	10	–	–	200	100	3
CSE-CIC-IDS2018	Confidence	10.0	–	–	–	–	–	0.0
CSE-CIC-IDS2018	ϵ	–	–	–	0.1	0.2	–	–
CSE-CIC-IDS2018	Learning rate	0.04	–	–	–	–	–	0.01
CSE-CIC-IDS2018	Step size	–	–	–	0.01	0.01	–	–
CSE-CIC-IDS2018	θ / γ	–	–	0.02 / 0.10	–	–	–	–
CSE-CIC-IDS2018	Maximum iterations	40	5	–	–	200	100	3

Computing environment and reproducibility. Experiments were executed on Linux with an NVIDIA T4 GPU (16 GB), 256 GB system memory, and a 48-core CPU. The implementation uses the Adversarial Robustness Toolbox [27], PyTorch, scikit-learn, XGBoost, and CatBoost. Results are point estimates from the fixed split used by the inherited experimental pipeline. No confidence intervals or repeated stochastic trials are available in the supplied artifacts.

5.2 Evasion and Transfer Attack Assessment

Performance of base learners under direct attacks. Table 5 reports direct black-box attacks for all components and DNN-originated white-box inputs for the tree targets. HSJA is the strongest direct query attack in both datasets: tree-target ASR ranges from 77.20% to 99.87% on NSL-KDD and from 69.68% to 90.86% on CSE-CIC-IDS2018, while direct DNN ASR reaches 100%. ZOO is strongly dataset dependent, producing substantial disruption on NSL-KDD but less than 0.12% ASR for every component on CSE-CIC-IDS2018 under the recorded configuration.

The direct DNN results also vary sharply by attack. DeepFool reaches 99.89% and 99.62% ASR, whereas JSMA and PGD record 0% ASR on the DNN despite nonzero effects on some transferred targets. These patterns should not be interpreted as intrinsic robustness to JSMA or PGD; they are configuration-specific outcomes and motivate the attack-audit limitations discussed later.

DNN-to-tree adversarial transfer. DeepFool exposes the strongest common-input DNN-to-tree pathway. DNN-generated examples produce tree-target ASRs of 69.25–80.17% on NSL-KDD and 49.35–88.40% on CSE-CIC-IDS2018. C&W transfers less uniformly, with 13.97–19.04% ASR on NSL-KDD and 25.37–51.18% on CSE-CIC-IDS2018. FGSM produces moderate transfer on NSL-KDD and weaker effects on most CSE-CIC-IDS2018 targets; JSMA is negligible, while PGD affects specific targets, notably BME on NSL-KDD and XGB on CSE-CIC-IDS2018.

Some target-side ASRs exceed the corresponding direct DNN ASR. This is consistent with Equation (6): the denominator is defined by the target’s initially correct set, and source failure is not a prerequisite. The measurement quantifies the delivered effect of a DNN-generated set on a tree target, not conditional transfer among source-successful examples.

Tree-to-DNN adversarial transfer. Table 6 evaluates the DNN on ZOO and HSJA examples generated against each tree source. On NSL-KDD, tree-originated ZOO yields 0.05–8.66% DNN

Table 5. Component performance under direct black-box attacks and DNN-originated white-box adversarial inputs (%). ZOO and HSJA are generated directly against each component. FGSM, JSMA, PGD, DeepFool, and C&W are generated against the DNN; tree entries are therefore transfer results.

Attack	Metric	XGB	CBT	GBM	BME	DNN
<i>NSL-KDD</i>						
Original	Acc	99.50	99.38	99.28	99.29	98.57
ZOO	Acc	42.92	44.09	66.49	59.62	1.79
	ASR	57.23	56.16	33.55	40.18	99.49
HSJA	Acc	3.89	22.97	13.14	0.55	1.15
	ASR	96.41	77.20	87.22	99.87	100.00
FGSM	Acc	82.57	75.76	74.88	64.14	87.59
	ASR	17.05	23.79	24.64	35.55	11.14
JSMA	Acc	99.63	99.54	99.46	99.35	99.81
	ASR	0.01	0.01	0.01	0.04	0.00
PGD	Acc	98.48	99.42	99.46	68.42	99.95
	ASR	1.39	0.39	0.01	31.32	0.00
DeepFool	Acc	30.76	24.19	26.49	19.88	0.87
	ASR	69.25	75.87	73.58	80.17	99.89
C&W	Acc	80.57	82.10	85.42	82.20	83.40
	ASR	19.04	17.38	13.97	17.25	15.39
<i>CSE-CIC-IDS2018</i>						
Original	Acc	95.95	95.90	95.91	95.91	93.39
ZOO	Acc	95.94	95.90	95.91	95.90	93.42
	ASR	0.07	0.12	0.01	0.05	0.08
HSJA	Acc	31.72	11.35	13.31	13.95	5.25
	ASR	69.68	90.86	88.77	87.61	100.00
FGSM	Acc	87.65	95.82	95.39	93.94	93.28
	ASR	9.02	0.29	0.62	2.17	0.11
JSMA	Acc	95.94	95.96	95.95	95.58	97.92
	ASR	0.14	0.11	0.18	0.68	0.00
PGD	Acc	74.20	95.97	95.78	89.61	93.71
	ASR	22.88	0.22	0.27	6.73	0.00
DeepFool	Acc	50.04	13.02	17.74	23.77	1.90
	ASR	49.35	88.40	82.58	76.19	99.62
C&W	Acc	69.97	47.36	71.90	52.20	32.47
	ASR	27.61	51.18	25.37	45.84	65.23

ASR and HSJA yields 0.29–21.38%, both far below the direct DNN-targeted references. On CSE-CIC-IDS2018, ZOO is ineffective in both direct and transferred conditions; HSJA transfers more strongly, reaching 31.04–62.54%, but remains below the 100% direct result.

Together with the DNN-originated results, the observations support attack-dependent operational directionality in the evaluated mixed-access setting. They do not establish architecture-only causality because the forward gradient attacks and reverse query attacks are not fully reciprocal, attack-matched experiments.

Table 6. DNN performance on black-box examples generated against tree sources (%). The DNN-source row is a direct-attack reference.

Generation source	ZOO		HSJA	
	Acc	ASR	Acc	ASR
<i>NSL-KDD</i>				
XGB	94.77	3.93	83.15	15.75
GBM	90.08	8.66	82.63	16.25
CBT	93.97	4.80	77.61	21.38
BME	98.54	0.05	98.36	0.29
DNN (direct)	1.79	99.49	1.15	100.00
<i>CSE-CIC-IDS2018</i>				
XGB	93.39	0.00	65.71	31.04
GBM	93.39	0.00	49.45	48.77
CBT	93.39	0.00	36.17	62.54
BME	93.39	0.00	62.94	34.43
DNN (direct)	93.42	0.08	5.25	100.00

Robustness of the undefended ensemble. Table 7 reports the fixed weighted baseline under the component-conditioned stress diagnostic. Voting absorbs attacks that leave most components correct: JSMA and PGD remain below 0.21% ASR on both datasets. It fails when family-conditioned inputs induce correlated errors. DeepFool reaches 98.60% ASR on NSL-KDD and 88.05% on CSE-CIC-IDS2018; HSJA reaches 61.93% and 68.98%, respectively. These values characterize the oracle-style stress condition, not a common-input physical-flow attack.

Robustness of HEDGE. Under the component-conditioned stress diagnostic, the complete HEDGE configuration markedly reduces attack success rates for the strongest attacks affecting the undefended ensemble. As shown in Table 8, DeepFool ASR decreases from 98.60% to 1.43% on NSL-KDD and from 88.05% to 2.03% on CSE-CIC-IDS2018. HSJA decreases from 61.93% to 0.11% and from 68.98% to 0.59%, while C&W decreases from 14.19% to 2.23% and from 46.36% to 0.45%, respectively. The complete configuration retains clean accuracies of 99.46% on NSL-KDD and 95.95% on CSE-CIC-IDS2018. These results address RQ3 by indicating that the defended heterogeneous ensemble preserves clean predictive utility while substantially reducing attack-induced error under the explicitly bounded stress protocol. The contributions of directional transfer-aware training, feature squeezing, and ensemble aggregation are further disentangled through the ablation analysis in Section 5.3.

5.3 Ablation Studies

Table 8 compares five matched configurations: Base denotes the undefended weighted ensemble; HEDGE-FS adds feature squeezing without adversarial retraining; HEDGE-IAT uses component-isolated adversarial training; HEDGE-DTAT replaces IAT with directional transfer-aware training without feature squeezing; and HEDGE denotes the complete configuration combining DTAT with the supporting inference mechanisms.

Isolated vs. Directional Transfer-Aware Training. The HEDGE-IAT–HEDGE-DTAT comparison isolates the added DNN-originated pool in the tree curricula. On NSL-KDD, DTAT reduces DeepFool ASR from 17.52% to 1.43%, FGSM from 2.01% to 0.21%, PGD from 0.84% to 0.08%, and

Table 7. Undefended ensemble under the component-conditioned stress diagnostic (%).

Attack	Acc	F1	Precision	Recall	ASR
<i>NSL-KDD</i>					
Original	99.46	99.45	99.45	99.46	–
ZOO	75.73	75.88	84.22	75.73	24.13
HSJA	38.15	40.04	48.51	38.15	61.93
FGSM	89.38	89.53	90.10	89.38	10.14
JSMA	99.70	99.70	99.70	99.70	0.01
PGD	99.93	99.93	99.93	99.93	0.01
DeepFool	1.60	1.74	1.60	1.60	98.60
C&W	85.35	85.49	87.02	85.35	14.19
<i>CSE-CIC-IDS2018</i>					
Original	96.01	96.00	96.05	96.01	–
ZOO	96.02	96.01	96.06	96.02	0.01
HSJA	32.20	27.51	60.72	32.20	68.98
FGSM	96.04	96.03	96.07	96.04	0.13
JSMA	96.26	96.25	96.32	96.26	0.07
PGD	95.98	95.98	96.02	95.98	0.21
DeepFool	13.09	7.93	6.65	13.09	88.05
C&W	51.87	47.74	75.69	51.87	46.36

C&W from 3.19% to 2.22%. On CSE-CIC-IDS2018, DeepFool decreases from 6.43% to 2.03%, FGSM from 0.27% to 0.07%, PGD from 0.19% to 0.08%, and C&W from 0.78% to 0.43%.

DTAT is not uniformly superior. HSJA changes from 0.11% to 0.15% on NSL-KDD and from 0.55% to 0.62% on CSE-CIC-IDS2018. The largest gains are concentrated on the DNN-originated optimization attacks represented by the added cross-family pool. This attack-aligned pattern provides the principal within-pipeline evidence for RQ2: under the fixed experimental realization, adding the selected DNN-originated pool is associated with lower ASR on several represented pathways, most prominently DeepFool, but not with a uniform improvement across all attacks.

Effect of feature squeezing. Feature squeezing alone is inconsistent. On NSL-KDD it reduces HSJA from 61.93% to 23.24% and FGSM from 10.14% to 5.61%, but has essentially no effect on DeepFool or C&W. On CSE-CIC-IDS2018, most changes are marginal. Once DTAT is active, HEDGE-DTAT and the complete HEDGE configuration differ by at most 0.04 percentage points for most attack–dataset pairs. Quantization should therefore be interpreted as a secondary stabilizer rather than the source of the principal gain.

Clean-data performance. All NSL-KDD variants retain 99.46% clean ensemble accuracy. On CSE-CIC-IDS2018, the undefended baseline achieves 96.01%, HEDGE-IAT 95.92%, HEDGE-DTAT 95.95%, and complete HEDGE 95.95%. The largest absolute difference is 0.09 percentage points. Because the results are fixed-split point estimates, these differences are not presented as statistically significant.

Ensemble contribution. The undefended ensemble improves over individual components only when attack-induced errors remain sufficiently uncorrelated. DTAT operates before aggregation by reducing errors along the added incoming pathway; voting then limits the influence of residual component failures. The current tables do not include a matched best-component-versus-ensemble

Table 8. ASR ablation under the component-conditioned stress diagnostic (%). Lower is better.

Attack	Base	HEDGE-FS	HEDGE-IAT	HEDGE-DTAT	HEDGE
<i>NSL-KDD</i>					
ZOO	24.13	22.74	0.10	0.09	0.10
HSJA	61.93	23.24	0.11	0.15	0.11
FGSM	10.14	5.61	2.01	0.21	0.24
JSMA	0.01	0.02	0.03	0.02	0.03
PGD	0.01	0.02	0.84	0.08	0.09
DeepFool	98.60	98.61	17.52	1.43	1.43
C&W	14.19	14.23	3.19	2.22	2.23
<i>CSE-CIC-IDS2018</i>					
ZOO	0.01	0.06	0.04	0.00	0.03
HSJA	68.98	67.64	0.55	0.62	0.59
FGSM	0.13	0.11	0.27	0.07	0.02
JSMA	0.07	0.11	0.11	0.06	0.07
PGD	0.21	0.23	0.19	0.08	0.07
DeepFool	88.05	88.05	6.43	2.03	2.03
C&W	46.36	46.40	0.78	0.43	0.45

Table 9. Contextual comparison of reported adversarial accuracy (%). Dashes denote unreported values; protocols are not necessarily matched.

Method	ZOO	HSJA	FGSM	JSMA	PGD	DeepFool	C&W
<i>NSL-KDD</i>							
HEDGE	99.40	99.37	99.39	99.58	99.65	98.21	97.26
APELID-ATH [10]	98.95	98.97	98.89	98.98	98.86	98.38	90.55
RF-RSE-AT [25]	–	–	72.90	–	71.38	–	–
Min–Max CNN [17]	–	–	95.69	–	95.52	94.31	92.67
Min–Max RNN [17]	–	–	95.53	–	95.52	95.32	95.21
DLL-IDS [44]	–	–	89.80	–	–	69.80	85.70
<i>CSE-CIC-IDS2018</i>							
HEDGE	95.94	95.79	96.93	97.33	98.45	94.90	95.75
Apollon [29]	93.04	75.50	–	–	–	–	–
Omni [35]	–	–	86.68	77.87	–	87.12	75.23
RAIDS [33]	–	–	–	–	59.00	–	–
DLL-IDS [44]	–	–	95.00	–	–	71.50	63.70

ablation after every training variant, so the marginal causal contribution of voting cannot be isolated as strongly as the IAT–DTAT effect. RQ3 is therefore answered conservatively: HEDGE preserves clean ensemble utility and substantially reduces the observed stress-test failures, while the evidence attributes the principal gain to adversarial retraining and the incremental directional gain to DTAT.

5.4 Contextual Comparison with State-of-the-Art Methods

Table 9 places HEDGE alongside representative adversarially robust IDS methods reported on NSL-KDD and CSE-CIC-IDS2018. HEDGE achieves adversarial accuracies of 97.26–99.65% on NSL-KDD and 94.90–98.45% on CSE-CIC-IDS2018 across the seven evaluated attacks. These results are broadly competitive with prior defenses while covering both gradient-/optimization-based attacks and query-based attacks. The comparison is not uniformly favorable to HEDGE; for example, APELID-ATH reports slightly higher accuracy under DeepFool on NSL-KDD. This avoids interpreting Table 9 as evidence of universal superiority.

Cross-paper values are not directly comparable because preprocessing, data splits, architectures, attack implementations, perturbation budgets, mutable-feature policies, and evaluation denominators differ across studies. Table 9 should therefore be interpreted as contextual positioning rather than a statistical ranking. Its main value is to show that HEDGE maintains competitive robustness over a relatively broad attack set, whereas several prior methods report results for only a subset of the considered attacks.

More importantly, the central evidence for DTAT does not come from cross-study differences, but from the controlled within-pipeline comparison in Table 8. Under otherwise matched settings, DTAT reduces DeepFool ASR relative to isolated adversarial training from 17.52% to 1.43% on NSL-KDD and from 6.43% to 2.03% on CSE-CIC-IDS2018, with additional gains for several DNN-originated attacks. Thus, Table 9 establishes competitive empirical positioning, whereas Table 8 provides the more direct evidence for the contribution of directional transfer-aware training.

5.5 Discussion

Security and deployment implications. The results support three bounded conclusions. First, architectural heterogeneity does not imply adversarial independence: a single DNN-originated feature vector can disrupt several tree targets under selected attacks. Second, the generation source matters for training. Direct query examples and transferred gradient-derived examples cover different empirical failure regions, and the latter materially reduce DeepFool exposure beyond isolated tree training. Third, inference transformations are not substitutes for model hardening; feature squeezing adds little after DTAT for most tested conditions.

A deployed heterogeneous IDS should therefore maintain a directed transfer profile rather than a single aggregate transfer score. Each material change to the feature pipeline, models, or traffic distribution can alter source–target relationships and should trigger re-evaluation. Every deployed component must receive a deterministic transformation of the same physical input. Family-specific adversarial inputs remain useful only as a diagnostic stress condition. Operationally, HEDGE shifts most cost to offline generation and retraining; the supplied experiments do not establish a latency, throughput, memory, or energy service-level guarantee.

Limitations and threats to validity. The strongest limitation is evidence scope. The attacks modify post-featurization vectors; type and range repair do not guarantee packet-level realizability or preserve every protocol and inter-feature dependency [1, 9]. ASR measures exact-label disruption, not strict malicious-to-benign bypass. The test sets are capped and do not reproduce deployment prevalence. The study covers two offline benchmarks, one DNN, and four tree learners, with no temporal split, campaign-separated validation, concept drift, or cross-dataset attack transfer.

Internal validity is also inherently bounded by our attack configurations. Training and evaluation use overlapping attack families, establishing attack-aware rather than zero-shot unseen-attack robustness. Furthermore, we rely on an established set of optimization and query-based attacks with fixed hyperparameters. Comprehensive robustness benchmarking additionally requires adaptive attack evaluation together with systematic auditing of attack objectives, budgets, optimization

settings, and failure modes [1, 2, 14]. Because feature quantization is not evaluated against an adaptive adversary that differentiates through the complete repair–quantization–ensemble pipeline using approximation techniques [2], our reported Attack Success Rates must be interpreted as *relative robustness gains* (isolating the marginal benefit of DTAT over isolated training) rather than absolute, worst-case guarantees. Additionally, forward and reverse directions use attack mechanisms compatible with different access modes and are not fully matched reciprocal experiments. The component-conditioned ensemble protocol can allocate different vectors to different branches and is not a realizable common-input attack. Finally, a single data split and the absence of repeated random seeds preclude formal statistical significance claims. The 0% PGD/JSMA and near-zero CSE-CIC-IDS2018 ZOO results require replication with objective-sign checks, random restarts, strict query accounting, and loss-progress diagnostics before supporting broad robustness conclusions.

6 Conclusions

This work reframed adversarial robustness in heterogeneous neural–tree intrusion-detection ensembles as a problem of directed incoming exposure rather than architectural diversity alone. By representing adversarial interactions as an ordered source–attack–target relation, we separated direct attacks, unchanged common-input transfer, and component-conditioned ensemble stress diagnostics, thereby making cross-model exposure an explicit object of analysis rather than an implicit consequence of ensemble construction.

Building on this formulation, HEDGE introduces Directional Transfer-Aware Training (DTAT), which preserves source–attack–target identity and uses selected adversarial pools to construct component-specific training curricula. The methodological contribution is therefore not transferred adversarial training itself, but an explicit threat-to-curriculum mapping that determines how source-indexed exposure is incorporated into heterogeneous ensemble components. The controlled evaluation supports this design principle: directional transfer is strongly attack and target dependent, and the additional robustness associated with DTAT is concentrated on incoming pathways represented in the transferred curriculum. These findings indicate that exposure-aware curriculum construction can complement component-isolated hardening without requiring architectural diversity to serve as a proxy for adversarial independence.

More broadly, the study suggests that heterogeneous ensemble robustness should be characterized not only by the standalone resilience of individual learners, but also by the directional relationships through which adversarial influence propagates across components. From this perspective, transferability is relevant not merely as an observed vulnerability or a criterion for model selection, but as information that can guide how robustness-oriented training data are assigned to deployed components.

A natural extension is to replace the current prespecified pathway policy with validation-driven allocation of adversarial-training capacity according to measured incoming transfer risk. This would move DTAT from fixed directional routing toward adaptive transfer-aware curriculum construction. Further work should also examine adaptive end-to-end attacks, held-out attack families, temporal and cross-dataset generalization, and problem-space realizability to determine how far the directional training principle extends beyond the controlled feature-space setting evaluated here.

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References

- [1] Giovanni Apruzzese, Mauro Andreolini, Luca Ferretti, Mirco Marchetti, and Michele Colajanni. 2022. Modeling Realistic Adversarial Attacks against Network Intrusion Detection Systems. *ACM Digital Threats: Research and Practice* 3, 3, Article 31 (2022), 19 pages. doi:10.1145/3469659
- [2] Anish Athalye, Nicholas Carlini, and David Wagner. 2018. Obfuscated Gradients Give a False Sense of Security: Circumventing Defenses to Adversarial Examples. In *Proceedings of the 35th International Conference on Machine Learning*. 274–283. doi:10.48550/arXiv.1802.00420
- [3] Zeinab Awad, Magdy Zakaria, and Rasha Hassan. 2025. An Enhanced Ensemble Defense Framework for Boosting Adversarial Robustness of Intrusion Detection Systems. *Scientific Reports* 15, Article 14177 (2025). doi:10.1038/s41598-025-94023-z
- [4] Vadim Borisov, Tobias Leemann, Kathrin Seßler, Johannes Haug, Martin Pawelczyk, and Gjergji Kasneci. 2024. Deep Neural Networks and Tabular Data: A Survey. *IEEE Transactions on Neural Networks and Learning Systems* 35, 6 (2024), 7499–7519. doi:10.1109/TNNLS.2022.3229161
- [5] Nicholas Carlini and David Wagner. 2017. Towards Evaluating the Robustness of Neural Networks. In *2017 IEEE Symposium on Security and Privacy*. 39–57. doi:10.1109/SP.2017.49
- [6] Jianbo Chen, Michael I. Jordan, and Martin J. Wainwright. 2020. HopSkipJumpAttack: A Query-Efficient Decision-Based Attack. In *2020 IEEE Symposium on Security and Privacy*. 1277–1294. doi:10.1109/SP40000.2020.00045
- [7] Pin-Yu Chen, Huan Zhang, Yash Sharma, Jinfeng Yi, and Cho-Jui Hsieh. 2017. ZOO: Zeroth Order Optimization Based Black-Box Attacks to Deep Neural Networks without Training Substitute Models. In *Proceedings of the 10th ACM Workshop on Artificial Intelligence and Security*. 15–26. doi:10.1145/3128572.3140448
- [8] Byung Chang Chung and Gyu-Bum Han. 2026. A DeepSHAP-Based Adversarial Attack on Machine Learning-Based Network Intrusion Detection. *IEEE Access* 14 (2026), 2566–2575. doi:10.1109/ACCESS.2025.3650123
- [9] Jacopo Cortellazzi, Erwin Quiring, Daniel Arp, Feargus Pendlebury, Fabio Pierazzi, and Lorenzo Cavallaro. 2025. Intriguing Properties of Adversarial ML Attacks in the Problem Space: Extended Version. *ACM Transactions on Privacy and Security* 28, 4, Article 42 (2025), 37 pages. doi:10.1145/3742895
- [10] Hanh P. Du, Minh Q. Tran, Tuyen T. Nguyen, and Hoa N. Nguyen. 2025. Multi-Layer Defense for AI-Powered IDS: Ensemble Adversarial Training and Explainable Resilience to Evasion Attacks. In *2025 2nd International Conference on Cryptography and Information Security*. 1–6. doi:10.1109/VCRIS68011.2025.11250565
- [11] Sabine Ennaji, Fabio de Gaspari, Dorjan Hitaj, Alicia Kbidi, and Luigi Vincenzo Mancini. 2025. Adversarial Challenges in Network Intrusion Detection Systems: Research Insights and Future Prospects. *IEEE Access* 13 (2025), 148613–148645. doi:10.1109/ACCESS.2025.3600984
- [12] Jalal Ghadermazi, Soumyadeep Hore, Ankit Shah, and Nathaniel D. Bastian. 2025. GTAE-IDS: Graph Transformer-Based Autoencoder Framework for Real-Time Network Intrusion Detection. *IEEE Transactions on Information Forensics and Security* 20 (2025), 4026–4041. doi:10.1109/TIFS.2025.3557741
- [13] Ian J. Goodfellow, Jonathon Shlens, and Christian Szegedy. 2015. Explaining and Harnessing Adversarial Examples. In *International Conference on Learning Representations*. doi:10.48550/arXiv.1412.6572
- [14] Dongqi Han, Zhiliang Wang, Ying Zhong, Wenqi Chen, Jiahai Yang, Shuqiang Lu, Xingang Shi, and Xia Yin. 2021. Evaluating and Improving Adversarial Robustness of Machine Learning-Based Network Intrusion Detectors. *IEEE Journal on Selected Areas in Communications* 39, 8 (2021), 2632–2647. doi:10.1109/JSAC.2021.3087242
- [15] M. S. Haroon and H. M. Ali. 2023. Ensemble Adversarial Training Based Defense against Adversarial Attacks for Machine Learning-Based Intrusion Detection System. *Neural Network World* 33, 5 (2023), 317–336. doi:10.14311/NNW.2023.33.018
- [16] Md Mehedi Hasan, Rafiqul Islam, Quazi Mamun, Md Zahid Islam, and Junbin Gao. 2026. Ensemble-Based Adversarial Defense with Dynamic Model Selection for Intrusion Detection Systems. *Internet of Things* 38, Article 101970 (2026), 22 pages. doi:10.1016/j.iot.2026.101970
- [17] Rana Abou Khamis and Ashraf Matrawy. 2024. Could Min–Max Optimization Be a General Defense against Adversarial Attacks?. In *2024 International Conference on Computing, Networking and Communications*. 671–676. doi:10.1109/ICNC59896.2024.10556266
- [18] Vishal Kumar, Krishan Kumar, Maninder Singh, and Neeraj Kumar. 2025. NIDS-DA: Detecting Functionally Preserved Adversarial Examples for Network Intrusion Detection System Using Deep Autoencoders. *Expert Systems with Applications* 270, Article 126513 (2025). doi:10.1016/j.eswa.2025.126513
- [19] Thi-Thu-Huong Le, Jaehan Cho, Dawit Shin, and Howon Kim. 2026. Toward Adversarial Robustness Network Intrusion Detection Based on Multi-Model Ensemble Approach. *Sensors* 26, 8, Article 2478 (2026). doi:10.3390/s26082478
- [20] Moshe Levy, Guy Amit, Yuval Elovici, and Yisroel Mirsky. 2024. Ranking the Transferability of Adversarial Examples. *ACM Transactions on Intelligent Systems and Technology* 15, 5, Article 100 (2024), 21 pages. doi:10.1145/3670409
- [21] Ying-Dar Lin, Jehoshua-Hanky Pratama, Didik Sudyana, Yuan-Cheng Lai, Ren-Hung Hwang, Po-Ching Lin, Hsuan-Yu Lin, Wei-Bin Lee, and Chen-Kuo Chiang. 2022. ELAT: Ensemble Learning with Adversarial Training in Defending against Evaded Intrusions. *Journal of Information Security and Applications* 71, Article 103348 (2022). doi:10.1016/j.jisa.

- 2022.103348
- [22] Mingqi Lv, Kehan Qian, Tieming Chen, Tiantian Zhu, and Jinyin Chen. 2025. ProvADShield: A Multimodel Ensemble Defender Against Adversarial Attacks on Provenance Graph Host Intrusion Detector. *IET Information Security*, Article 8625988 (2025), 21 pages. doi:10.1049/ise2/8625988
 - [23] Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu. 2018. Towards Deep Learning Models Resistant to Adversarial Attacks. In *International Conference on Learning Representations*. doi:10.48550/arXiv.1706.06083
 - [24] Seyed-Mohsen Moosavi-Dezfooli, Alhussein Fawzi, and Pascal Frossard. 2016. DeepFool: A Simple and Accurate Method to Fool Deep Neural Networks. In *2016 IEEE Conference on Computer Vision and Pattern Recognition*. 2574–2582. doi:10.1109/CVPR.2016.282
 - [25] D. Nathaniel and A. Soosai. 2023. A Robust Network Intrusion Detection System Using Random Forest Based Random Subspace Ensemble to Defend against Adversarial Attacks. *Advances in Electrical and Computer Engineering* 23, 4 (2023), 81–88. doi:10.4316/AECE.2023.04009
 - [26] Tuyen T. Nguyen, Uyen H. Tran, and Hoa N. Nguyen. 2025. Adversarial Attack and Defense in AI-Powered Intrusion Detection. *Journal of Computer Science and Cybernetics* 41, 4 (2025), 387–402. doi:10.15625/1813-9663/22884
 - [27] Maria-Irina Nicolae, Mathieu Sinn, Minh Ngoc Tran, Beat Buesser, Ambrish Rawat, Martin Wistuba, Valentina Zantedeschi, Nathalie Baracaldo, Bryant Chen, Heiko Ludwig, Ian Molloy, and Ben Edwards. 2018. Adversarial Robustness Toolbox v1.0.0. *arXiv preprint arXiv:1807.01069* (2018). doi:10.48550/arXiv.1807.01069
 - [28] Nicolas Papernot, Patrick McDaniel, Somesh Jha, Matt Fredrikson, Z. Berkay Celik, and Ananthram Swami. 2016. The Limitations of Deep Learning in Adversarial Settings. In *2016 IEEE European Symposium on Security and Privacy*. 372–387. doi:10.1109/EuroSP.2016.36
 - [29] Antonio Payá, Sergio Arroni, Vicente García-Díaz, and Alberto Gómez. 2024. Apollon: A Robust Defense System against Adversarial Machine Learning Attacks in Intrusion Detection Systems. *Computers & Security* 136, Article 103546 (2024). doi:10.1016/j.cose.2023.103546
 - [30] Noora Al Roken, Hakim Hacid, Ahmed Bouridane, and Abir Hussain. 2025. On Adversarial Attack Detection in the Artificial Intelligence Era: Fundamentals, a Taxonomy, and a Review. *Intelligent Systems with Applications* 27, Article 200554 (2025). doi:10.1016/j.iswa.2025.200554
 - [31] Noora Al Roken, Hakim Hacid, Ahmed Bouridane, and Abir Hussain. 2026. Hybrid Adversarial Retraining Approach for a Proactive Network Intrusion Detection System. *IEEE Open Journal of the Communications Society* 7 (2026), 5350–5364. doi:10.1109/OJCOMS.2026.3694298
 - [32] Ishai Rosenberg, Asaf Shabtai, Yuval Elovici, and Lior Rokach. 2021. Adversarial Machine Learning Attacks and Defense Methods in the Cyber Security Domain. *Comput. Surveys* 54, 5, Article 108 (2021), 36 pages. doi:10.1145/3453158
 - [33] Alper Sarıkaya, Banu Günel Kılıç, and Mehmet Demirci. 2023. RAIDS: Robust Autoencoder-Based Intrusion Detection System Model against Adversarial Attacks. *Computers & Security* 135, Article 103483 (2023). doi:10.1016/j.cose.2023.103483
 - [34] Iman Sharafaldin, Arash Habibi Lashkari, and Ali A. Ghorbani. 2018. Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization. In *Proceedings of the 4th International Conference on Information Systems Security and Privacy*. 108–116. doi:10.5220/0006639801080116
 - [35] Rui Shu, Tianpei Xia, Laurie Williams, and Tim Menzies. 2022. Omni: Automated Ensemble with Unexpected Models against Adversarial Evasion Attack. *Empirical Software Engineering* 27, 1, Article 28 (2022). doi:10.1007/s10664-021-10064-8
 - [36] Mahbod Tavallae, Ebrahim Bagheri, Wei Lu, and Ali A. Ghorbani. 2009. A Detailed Analysis of the KDD CUP 99 Data Set. In *2009 IEEE Symposium on Computational Intelligence for Security and Defense Applications*. 1–6. doi:10.1109/CISDA.2009.5356528
 - [37] Florian Tramèr, Alexey Kurakin, Nicolas Papernot, Ian Goodfellow, Dan Boneh, and Patrick McDaniel. 2018. Ensemble Adversarial Training: Attacks and Defenses. In *International Conference on Learning Representations*. doi:10.48550/arXiv.1705.07204
 - [38] Hoang V. Vo, Hanh P. Du, and Hoa N. Nguyen. 2023. AI-Powered Intrusion Detection in Large-Scale Traffic Networks Based on Flow Sensing Strategy and Parallel Deep Analysis. *Journal of Network and Computer Applications* 220, Article 103735 (2023). doi:10.1016/j.jnca.2023.103735
 - [39] Hoang V. Vo, Hanh P. Du, and Hoa N. Nguyen. 2024. APELID: Enhancing Real-Time Intrusion Detection with Augmented WGAN and Parallel Ensemble Learning. *Computers & Security* 136, Article 103567 (2024). doi:10.1016/j.cose.2023.103567
 - [40] Minxiao Wang, Ning Yang, Nicolas J. Forcade-Perkins, and Ning Weng. 2024. ProGen: Projection-Based Adversarial Attack Generation Against Network Intrusion Detection. *IEEE Transactions on Information Forensics and Security* 19 (2024), 5476–5491. doi:10.1109/TIFS.2024.3402155

- [41] Ning Wang, Yuchen Chen, Yuwei Hu, Wenjing Lou, and Y. Thomas Hou. 2021. MANDA: On Adversarial Example Detection for Network Intrusion Detection System. *IEEE INFOCOM 2021 - IEEE Conference on Computer Communications* (2021), 1–10. doi:10.1109/INFOCOM42981.2021.9488874
- [42] Cong Wu, Jianfei Sun, Jing Chen, Mamoun Alazab, Yang Liu, and Yang Xiang. 2025. TCG-IDS: Robust Network Intrusion Detection via Temporal Contrastive Graph Learning. *IEEE Transactions on Information Forensics and Security* 20 (2025), 1475–1486. doi:10.1109/TIFS.2025.3530702
- [43] Weilin Xu, David Evans, and Yanjun Qi. 2018. Feature Squeezing: Detecting Adversarial Examples in Deep Neural Networks. In *Proceedings of the Network and Distributed System Security Symposium*. doi:10.14722/ndss.2018.23198
- [44] Xinwei Yuan, Shu Han, Wei Huang, Hongliang Ye, Xianglong Kong, and Fan Zhang. 2024. A Simple Framework to Enhance the Adversarial Robustness of Deep Learning-Based Intrusion Detection System. *Computers & Security* 137, Article 103644 (2024). doi:10.1016/j.cose.2023.103644
- [45] Chaoyun Zhang, Xavier Costa-Pérez, and Paul Patras. 2022. Adversarial Attacks Against Deep Learning-Based Network Intrusion Detection Systems and Defense Mechanisms. *IEEE/ACM Transactions on Networking* 30, 3 (2022), 1294–1311. doi:10.1109/TNET.2021.3137084
- [46] Hangsheng Zhang, Dongqi Han, Shangyuan Zhuang, Zhiliang Wang, Jiyan Sun, Yinlong Liu, Jiqiang Liu, and Jinsong Dong. 2025. Explainable and Transferable Adversarial Attack for ML-Based Network Intrusion Detectors. *IEEE Transactions on Dependable and Secure Computing* 22, 5 (2025), 5090–5107. doi:10.1109/TDSC.2025.3560486
- [47] Yihang Zhang, Yingwen Wu, and Xiaolin Huang. 2024. Toward Transferable Adversarial Attacks Against Autoencoder-Based Network Intrusion Detectors. *IEEE Transactions on Industrial Informatics* 20, 12 (2024), 13863–13872. doi:10.1109/TII.2024.3435532